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UPTD-net: Unsupervised polarization-guided two-stage decoupling for underwater low-light image restoration

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HIGHLIGHTS

- An unsupervised learning method achieves underwater low-light image restoration.
- A novel imaging model jointly formulates scattering and illumination degradation.
- Polarization serves as a dual physical prior to guide the decoupling process.

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ABSTRACT

Underwater images captured under low-light conditions suffer from the coupled effects of scattering and illumination degradation, which reduce contrast, obscure details, and limit visibility. However, existing polarization-based methods often rely on accurate parameter estimation, making restoration sensitive to model assumptions and parameter errors. Supervised methods require paired datasets that are difficult to acquire in real underwater environments. To address these limitations, we formulate an underwater low-light imaging model inspired by Retinex theory and propose UPTD-Net, an unsupervised two-stage decoupling framework that optimizes each input without paired data or pretraining. UPTD-Net decomposes the coupled degradation into scattering removal and illumination recovery. Rather than using physical parameters estimated from polarization to directly determine the restored image, it incorporates polarization information into both stages as spatial weights for descattering refinement and a soft target for degraded illumination estimation. This design incorporates polarization information throughout restoration while reducing reliance on precise physical parameter estimates. Extensive experiments demonstrate that the proposed framework effectively restores underwater images under complex degradation conditions, achieving favorable quantitative performance and visual quality. The code and datasets will be available at https://github.com/fxt-kby/UPTD-Net.

1. Introduction

Underwater imaging plays a vital role in marine exploration, archaeology, and biological research [1–4]. However, in complex underwater environments, light scattering causes degradation of image contrast and blurring of details [5–7], while light absorption results in severe color distortion and low illumination [8–10]. These two degradations are coupled and interact with each other, making underwater image enhancement (UIE) a challenging task.

Under low-light conditions, such as in the deep sea or at night, these degradation problems become more serious. Images then exhibit degraded contrast and color, and become dark with poor visibility. The prior assumptions of traditional physics-based methods [11–13] often fail in low-light environments, leading to incorrect parameter estimation and consequently amplifying noise. Despite the considerable progress in learning-based underwater image enhancement [14–17], supervised learning-based methods require paired training data, which are

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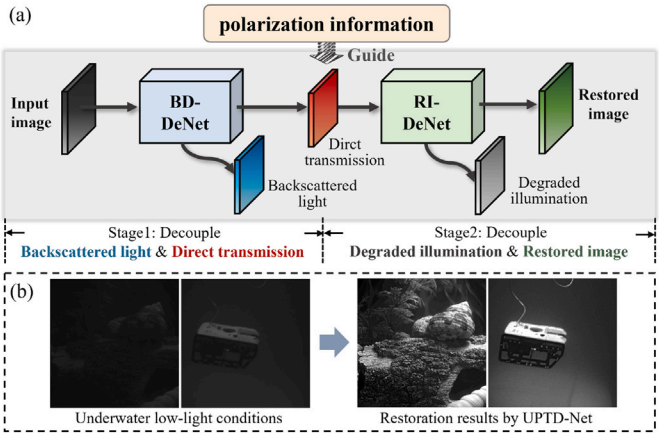


Fig. 1. The proposed two-stage decoupling framework. (a) The proposed two-stage framework: BD-DeNet removes scattering by decoupling backscattered light from direct transmission, and RI-DeNet recovers illumination by decoupling degraded illumination from object reflectance. Polarization information guides both stages. (b) Restored results using our framework.

difficult to obtain in real underwater environments. Furthermore, when addressing complex degradations caused by multiple factors, end-to-end strategies often result in limited performance. Therefore, the question of how to achieve effective decoupling of scattering and illumination degradation from a physical imaging model, without the need for paired data, is a core problem that needs to be solved.

Since backscattered light is partially polarized [18], processing the polarization information can effectively suppress scattered light. Polarization cameras, such as the Lucid Vision Phoenix, can easily capture multiple polarization images in a single shot, facilitating the development of polarization imaging. Polarization parameters including the degree of polarization (DoP), angle of polarization (AoP), and Stokes vector have been widely used in polarization-based descattering tasks [19–21]. Physics-based polarization methods generally estimate scattering-related parameters from polarization observations and recover scene information through physical models [12,18,19,22]. Under low-light conditions, weak signals and sensor noise will reduce the accuracy of these estimates and affect restoration results. Learning-based methods commonly use multi-angle polarization images as network inputs or infer physical variables, while supervised methods require paired polarization data for training [17,23,24]. In contrast, the proposed method incorporates polarization information as a reference for image refinement and a soft constraint for illumination estimation within an unsupervised optimization framework to improve restoration robustness.

This paper introduces a novel underwater low-light imaging model that jointly considers image degradation caused by light scattering and low illumination. The proposed UPTD-Net decomposes the degradation process into two physically clearer and more tractable subtasks: scattering removal and illumination recovery, as shown in Fig. 1. Since each sub-network deals with only one degradation, the structures of these two sub-networks can be simple. In addition, we introduce polarization information as a dual physical prior that is deeply involved in both subtasks, enabling natural illumination recovery and detail preservation. During restoration, UPTD-Net is optimized independently for each input polarization image set in an unsupervised manner. It effectively restores underwater low-light images and generalizes well to non-polarized underwater and hazy images. The contributions of this paper can be summarized as follows.

1. We reformulate the underwater low-light imaging model and propose the UPTD-Net framework to decompose the complex restoration into two cascaded decoupling tasks: scattering removal and illumination recovery, which are trained without paired datasets.
2. We introduce polarization information as a global dual physical prior to guide both descattering optimization and illumination recovery, providing a new application approach for underwater polarization imaging.
3. Comprehensive experiments demonstrate that UPTD-Net achieves favorable quantitative and visual performance, with additional generalization results on color polarized images, non-polarized underwater images, and hazy images.

2. Related work

This section reviews polarization imaging models and UIE methods, including physical model-based, image enhancement -based, and deep learning-based methods.

2.1. Polarization imaging model

Stokes vectors can comprehensively describe the polarization state of light and serve as a fundamental descriptor in polarization imaging [12,19]. Neglecting circular polarization, Stokes vectors can be calculated from four polarization images $I_{0^\circ}, I_{45^\circ}, I_{90^\circ}$ and I_{135°, as follows:

$$\begin{bmatrix} S_0 \\ S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\left(I_{0^\circ} + I_{45^\circ} + I_{90^\circ} + I_{135^\circ}\right) \\ I_{0^\circ} - I_{90^\circ} \\ I_{45^\circ} - I_{135^\circ} \end{bmatrix}, \tag{1}$$

where S_0, S_1, and S_2 are Stokes parameters. The degree of polarization DoP can be defined as follows:

$$DoP = \frac{\sqrt{S_1^2 + S_2^2}}{S_0}. \tag{2}$$

The Schechner model [18] exploits the partial polarization of backscattered light for descattering and assumes that direct transmission is unpolarized. The parallel and perpendicular polarization components, denoted as $S^{\parallel}(x, y)$ and $S^{\perp}(x, y)$, respectively, can be expressed as:

$$\begin{aligned} S^{\parallel}(x, y) &= \frac{D(x, y)}{2} + B^{\parallel}(x, y), \\ S^{\perp}(x, y) &= \frac{D(x, y)}{2} + B^{\perp}(x, y), \end{aligned} \tag{3}$$

where $D(x, y)/2$ is the intensity of direct transmission through the polarizer. $B^{\parallel}(x, y)$ and $B^{\perp}(x, y)$ are the parallel and perpendicular components of the backscattered light. The degree of polarization DoP and the degree of polarization of the backscattered light DoP_B can be defined as follows:

$$\begin{aligned} DoP(x, y) &= \frac{S^{\parallel}(x, y) - S^{\perp}(x, y)}{S^{\parallel}(x, y) + S^{\perp}(x, y)}, \\ DoP_B(x, y) &= \frac{B^{\parallel}(x, y) - B^{\perp}(x, y)}{B^{\parallel}(x, y) + B^{\perp}(x, y)}. \end{aligned} \tag{4}$$

These relationships provide a physical basis for using polarization information for underwater image restoration.

2.2. Underwater image enhancement methods

1) *Physical Model-Based Methods*: Model-based methods model the underwater image degradation process by integrating physical principles with image priors. Various priors have been proposed, including the dark channel prior (DCP) [11] and its derivatives [25], [26], the red channel prior (RCP) [8], the haze-lines model [9], the histogram distribution prior [6], the maximum intensity prior (MIP) [27], and

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the underwater light attenuation prior (ULAP) [28]. Hou et al. [29] proposed an illumination channel sparsity prior for non-uniform underwater illumination. Zhang et al. [30] divided the image into blocks and designed locally adaptive color correction and contrast enhancement to restore underwater images.

The difference in polarization between object and scattered light gives polarization imaging a distinct advantage in underwater vision [18,22]. Underwater polarization imaging methods are mainly based on polarization difference imaging [18,31], the Stokes vector [12, 22], or the Mueller matrix [32]. Wei et al. [33] made full use of polarization information (including DoP and AoP) from the Stokes vector to remove scattering and enhance underwater vision. Chen et al. [19] proposed the neighborhood high fidelity constraint to optimize the estimation of the DoP of background light, thereby improving the performance. Li et al. [3] further exploited the angle of polarization with low-rank and sparse decomposition for robust underwater polarization image recovery.

Although these models provide a physical foundation for understanding underwater degradation, their performance is highly dependent on the validity of these priors. Such methods are often effective only for specific degradation and may fail under complex conditions.

2) *Image Enhancement-Based Methods*: These methods directly improve image contrast and color by redistributing pixel intensities. Key techniques include histogram equalization (HE) [5], contrast-limited adaptive histogram equalization (CLAHE) [7], global histogram stretching [34], Retinex-based methods [35], and the probability-based method (PB) [36]. Zhang et al. [37] adopted a weighted wavelet fusion strategy to enhance underwater images. Zheng et al. [38] proposed an adaptive threshold segmentation and histogram equalization method to improve highly turbid underwater images.

Image enhancement-based methods directly manipulate the pixel distributions, without considering the fundamental physics of imaging. Consequently, in low-light underwater environments, this approach tends to severely amplify image background noise while causing unnatural color shifts.

3) *Deep Learning-Based Methods*: Deep learning methods have become increasingly important in underwater image enhancement, ranging from end-to-end data-driven restoration to hybrid methods that combine learning architectures with physical imaging models. Representative methods include CNN-based models, such as PUIE [16] and UICoE-Net [39]; attention-based models, such as WaveNet [40] and HAAM-GAN [41]; and Transformer-based architectures including U-Trans [15] and UWAGA [42]. Frequency-domain information has been exploited through Fourier-transform-based methods [43,44] and wavelet-decomposition-based methods [45]. Diffusion-based methods have also been introduced into underwater image enhancement, such as DiffWater [46], WF-Diff [47], and WDUIE [48].

Learning-based underwater polarization imaging methods can be broadly categorized into CNN-based [17,23], GAN-based [24], Transformer-based [49,50], and diffusion-based methods [51]. However, supervised learning approaches are limited by the difficulty of acquiring paired training data [15,52]. Although unsupervised frameworks address this constraint, many still rely on end-to-end mappings and thus exhibit unstable performance and limited interpretability under coupled degradations. To overcome these challenges, we propose a two-stage decoupling method that explicitly models the degradation process, establishing an effective and interpretable framework for underwater image restoration.

3. Proposed method

The architecture of our proposed UPTD-Net consists of two stages: the backscattered light and direct transmission decoupling network (BD-DeNet) and the reflectance and degraded illumination decoupling network (RI-DeNet), as shown in Fig. 2. Polarization priors are incorporated across the overall framework. Specifically, the perpendicular polarization component is used to refine the descattered image, while DoP provides a soft reference for degraded illumination estimation.

The remainder of this section is organized as follows: Section 3.1 introduces our proposed underwater low-light imaging model. Section 3.2 describes the decoupling of backscattered light and direct transmission using BD-DeNet. Section 3.3 presents the polarization-guided descattering optimization component. Finally, Section 3.4 introduces the decoupling of reflectance and degraded illumination via RI-DeNet.

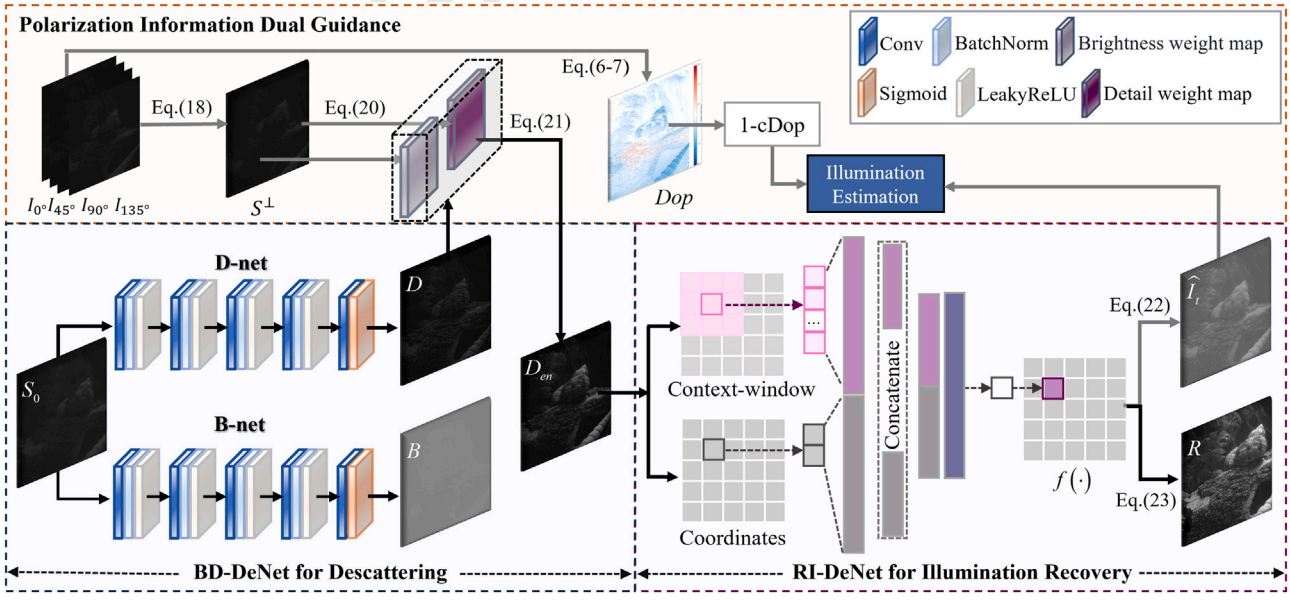


Fig. 2. Proposed UPTD-Net algorithm flow. UPTD-Net uses dual polarization guidance in two stages. (1) BD-DeNet for descattering stage: D-Net and B-Net extract direct transmission D and backscattered light B, respectively. Using $S^{\perp}$ as a reference, the polarization-guided descattering optimization module extracts brightness and detail weight maps to enhance D, resulting in D_{en}. (2) RI-DeNet for illumination recovery stage: guided by the polarization prior $(1 - cDoP)$ for illumination estimation, it combines context-window and coordinates to decompose D_{en} into degraded illumination $\hat{I}_t$ and restored image R.

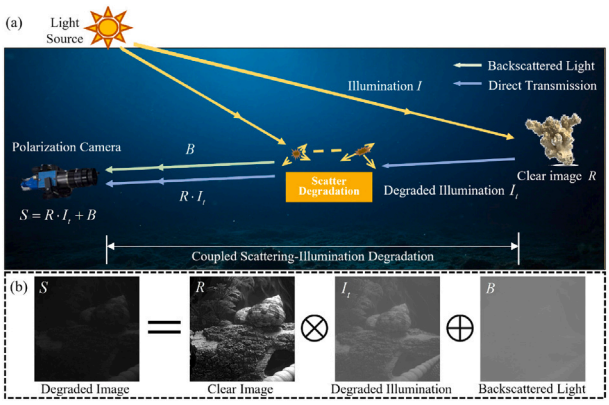


Fig. 3. Underwater low-light imaging model. (a) The polarization camera captures the degraded image $S = R \cdot I_t + B$ under coupled scattering and illumination degradation. (b) S is decomposed into clear image R, degraded illumination I_t and backscattered light B, where $\oplus$ and $\otimes$ denote addition and multiplication.

3.1. Underwater low-light imaging model

The underwater image formation process can be described by the Jaffe–McGlamery model [53,54]:

$$\begin{aligned} S(x,y) &= D(x,y) + B(x,y) \\ &= J(x,y)t(x,y) + B_\infty(1 - t(x,y)), \end{aligned} \tag{5}$$

where $S(x,y)$ is the observed image, $D(x,y)$ and $B(x,y)$ denote the direct transmission and backscattered light, respectively, $J(x,y)$ is the scene radiance, $t(x,y)$ is the transmission map, and B_∞ is the background light. From Eq. (5), the transmission can be expressed as:

$$t(x,y) = 1 - \frac{B(x,y)}{B_\infty}. \tag{6}$$

To model illumination degradation under low-light conditions, we introduce the Retinex decomposition [55],

$$J(x,y) = I(x,y)R(x,y). \tag{7}$$

where $I(x,y)$ and $R(x,y)$ denote the illumination and reflectance, respectively.

The observed image captured by the detector can be modeled by combining Eqs. (5) and (7) as follows:

$$S(x,y) = I(x,y)R(x,y)t(x,y) + B(x,y). \tag{8}$$

In our formulation, $R(x,y)$ represents the clear image without scattering and illumination degradation. Since illumination $I(x,y)$ and transmission $t(x,y)$ jointly modulate the direct component in a multiplicative form, we combine them into a degraded illumination component $I_t(x,y) = I(x,y)t(x,y)$, as shown in Fig. 3. This design reduces the number of unknown variables and alleviates error propagation caused by separately estimating the transmission map. Thus, Eq. (8) can be rewritten as follows:

$$S(x,y) = R(x,y)I_t(x,y) + B(x,y). \tag{9}$$

Based on Eq. (9), the restoration problem is decomposed into two cascaded stages. First, BD-DeNet decouples the observed image $S(x,y)$ into $R(x,y)I_t(x,y)$ and $B(x,y)$. Then, RI-DeNet separates direct transmission into $R(x,y)$ and $I_t(x,y)$. The polarization-guided descattering optimization component is designed to enhance the final restoration quality.

3.2. BD-DeNet for descattering

We propose BD-DeNet to decouple direct transmission and backscattered light for descattering. Its dual-branch design follows their distinct physical characteristics: direct transmission is rich in high-frequency scene details, whereas backscattered light exhibits low-frequency and spatially smooth characteristics [56,57]. Accordingly, D-Net and B-Net are used to estimate the direct transmission D and backscattered light B, respectively.

1) *D-Net*: D-Net estimates direct transmission using a multi-scale convolutional structure, as shown in Fig. 2. The network is composed of five convolutional blocks. The first four blocks each contain a convolutional layer, batch normalization, and LeakyReLU activation. The final block serves as the output layer and consists of a convolutional layer followed by a sigmoid function. A progressive kernel expansion strategy is adopted to capture multi-scale features, especially complex edges and textures in underwater scenes.

Without paired training data, we introduce several constraints to optimize D-Net. The total loss $\mathcal{L}_D$ is defined as follows:

$$\mathcal{L}_D = \lambda_1 \mathcal{L}_{MSE} + \lambda_2 \mathcal{L}_{grad} + \lambda_3 \mathcal{L}_{rec}, \tag{10}$$

where $\mathcal{L}_{MSE}$, $\mathcal{L}_{grad}$ and $\mathcal{L}_{rec}$ represent fidelity, gradient loss and reconstruction loss, respectively. λ_1, λ_2 and λ_3 are trade-off parameters.

To maintain pixel-level consistency between the estimated $D(x,y)$ and the observed image $S(x,y)$, we use mean squared error (MSE) for the fidelity loss $\mathcal{L}_{MSE}$, as follows:

$$\mathcal{L}_{MSE} = \frac{1}{HW}\|S - D\|_2^2, \tag{11}$$

where $H \times W$ is the image size of S and D, and $\|\cdot\|_2$ denotes the L2 norm.

To ensure the direct transmission D has rich edge and texture information, we introduce gradient loss $\mathcal{L}_{grad}$ as follows:

$$\mathcal{L}_{grad} = \|\nabla D - \nabla S\|_2^2, \tag{12}$$

where ∇ is the gradient operator.

Since D-Net and B-Net form a dual-branch parallel network, their outputs should satisfy the physical constraint in (5). Therefore, the reconstruction loss $\mathcal{L}_{rec}$ is introduced to enforce consistency with the underwater imaging model:

$$\mathcal{L}_{rec} = \|(D + B) - S\|_2^2. \tag{13}$$

2) *B-Net*: The backscattered light estimation network B-Net is composed of five convolutional blocks, similar in structure to D-Net. However, unlike D-Net, all convolutional layers in B-Net use kernels of the same size. Due to the sparse details in B, this simplified design can improve computational efficiency.

For B-Net, we also design loss functions to guide the network in learning features of backscattered light and to promote stable training. The total loss for B-Net, denoted as $\mathcal{L}_B$, is defined as follows:

$$\mathcal{L}_B = \lambda_4 \mathcal{L}_{low} + \lambda_5 \mathcal{L}_{TV} + \lambda_6 \mathcal{L}_{rec}, \tag{14}$$

where $\mathcal{L}_{low}$ and $\mathcal{L}_{TV}$ denote the low-frequency and smoothness constraints, respectively. $\mathcal{L}_{rec}$ is the reconstruction loss, same as in D-Net. λ_4, λ_5 and λ_6 are trade-off parameters.

Due to water scattering in (5), the backscattered light B shows low-frequency characteristics, which inspired many methods to use low-pass filtered images as representations of B [56,57]. Therefore, we introduce $\mathcal{L}_{low}$ to guide B-Net toward a low-frequency representation, as follows:

$$\mathcal{L}_{low} = \|B - S_{\text{low}}\|_2^2, \tag{15}$$

where $S_{\text{low}} = \mathcal{F}_{\text{LP}}(S)$ is the Gaussian low-pass filtered image.

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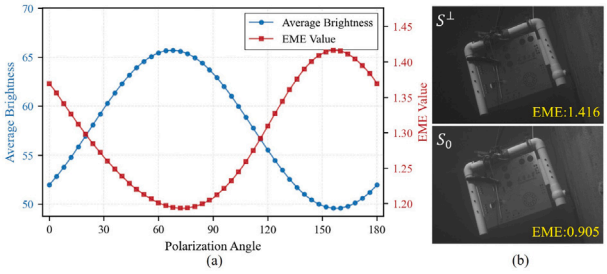


Fig. 4. Relationship between image brightness and EME value. (a) average brightness (blue line) and EME value (red line) varying with polarization angle, showing a negative correlation between brightness and information richness. (b) comparison showing that $S^{\perp}$ has higher EME value (1.416) than S_0 (0.905) despite lower brightness.

Smoothness is another characteristic of backscattered light, which is enforced by the total variation (TV) as follows:

$$\mathcal{L}_{TV} = \left\|\nabla_x B\right\|_2^2 + \left\|\nabla_y B\right\|_2^2, \tag{16}$$

where ∇_x and ∇_y represent the horizontal and vertical gradient operations, respectively. Here, the low-frequency property of backscattered light serves as a soft prior. BD-DeNet combines low-frequency, smoothness, reconstruction, and gradient constraints to reduce decomposition ambiguity and suppress structure leakage into B.

BD-DeNet decouples the observed image S, resulting in an initial descattered result D. However, D still suffers from loss of detail and insufficient brightness.

3.3. Polarization-guided descattering optimization (PDO)

Based on the Stokes representation, the polarization image $I_{pol}(\theta)$ at any angle θ can be expressed as:

$$\begin{aligned} I_{pol}(\theta) &= \frac{1}{2}\left[S_0 + S_1\cos(2\theta) + S_2\sin(2\theta)\right] \\ &= \frac{1}{2}\left[S_0 + \sqrt{S_1^2 + S_2^2}\cos(2\theta - \varphi)\right], \end{aligned} \tag{17}$$

where θ is the angle of polarization, and $\varphi = \arctan(S_2/S_1)$.

By combining Eqs. (4) and (2), (17) can be expressed as the sum of directly transmitted light and a function of backscattered light, as follows:

$$I_{pol}(\theta) = \frac{D}{2} + \frac{B}{2}\left[1 + DoP_B\cos(2\theta - \varphi)\right]. \tag{18}$$

When $\cos(2\theta - \varphi) = -1$, backscattered light is minimized, resulting in the lowest image brightness but the richest details, which correspond to the perpendicular polarization component $S^{\perp}(x, y) = 1/2 * \left[S_0 - \sqrt{S_1^2 + S_2^2}\right]$.

We use the Enhancement Measure Evaluation (EME) [58] to assess image information richness. Experiments on low-light underwater images reveal a negative correlation between EME and average brightness, as shown in Fig. 4. For images with lower average brightness, only a small amount of light reaches the detector, resulting in less scattering and preserving richer image information. As shown in Fig. 4(b), although $S^{\perp}(x, y)$ is darker than the original intensity image, its EME value is higher than that of S_0. Thus, we use $S^{\perp}(x, y)$ as the reference to optimize the descattered image.

Based on $S^{\perp}(x, y)$, we construct a detail weight map and a brightness weight map to refine the initial descattered result D. The detail weight map W_d is generated through weighted fusion of gradient features and local entropy features, which identifies regions that need enhancement, as follows:

$$W_d = \frac{\bar{W}_g}{\bar{W}_g + \bar{W}_e} W_g + \frac{\bar{W}_e}{\bar{W}_g + \bar{W}_e} W_e, \tag{19}$$

where W_d is composed of gradient feature map W_g computed via the Sobel operator and local entropy feature map W_e. $\bar{W}_g$ and $\bar{W}_e$ are the mean values of the respective feature maps.

The brightness weight map $W_b(x, y)$ is obtained by normalizing $S^{\perp}(x, y)$ to [0,1], thereby preserving its relative intensity distribution. Laplacian pyramid enables multiscale image representation for processing details at various scales. We use this representation for multiscale detail enhancement, where each pyramid level L_l is enhanced as $L_l' = L_l \cdot (1 + W_d)$.

Finally, we optimize the initial descattered result D obtained from BD-DeNet, and the overall process of detail and brightness optimization can be formulated as follows:

$$D_{en} = \left(G_3 + \sum_{l=0}^{2} L_l'\right) + \eta \cdot W_b \cdot (1 - D), \tag{20}$$

where D_{en} denotes the enhanced result, G_3 is the top level of the Gaussian pyramid, L_l' represents the enhanced Laplacian pyramid levels ($l = 0, 1, 2$), and $\eta = 0.8$ controls the brightness optimization. The term $(1 - D)$ serves as an adaptive saturation margin to prevent overexposure while enhancing dark regions.

PDO constructs the detail and brightness weight maps from the perpendicular polarization component and uses them to refine D to obtain D_{en}. It contains no trainable parameters and requires no separate optimization. The resulting D_{en} is then processed by RI-DeNet to estimate the reflectance R and degraded illumination I_t according to Eq. (9).

3.4. RI-DeNet for illumination recovery

RI-DeNet employs neural implicit representation (NIR) [59,60] to model spatially varying degraded illumination and to decouple it from object reflectance. It processes information from D_{en}, as shown in Fig. 2. The input is divided into two streams: one for pixel coordinates and the other for local context windows. Each stream is processed by a dedicated branch with 256 hidden channels. Before merging, the feature dimension of each branch is reduced by half. The compressed features are then concatenated to fuse spatial and contextual information and then fed into the output layers. This design enables the network to learn an implicit representation that captures both spatial relationships and local contextual information.

According to the underwater low-light imaging model in (9), we have $D_{en} = R \cdot I_t$. Since illumination and low-light observations are similar and exhibit linear relationship in most regions, a residual function $f(\cdot)$ is introduced to learn the residual between D_{en} and I_t. This reduces computational complexity while improving the robustness of illumination estimation. The estimated illumination $\hat{I}_t$ is given by:

$$\hat{I}_t = D_{en} + f(\cdot). \tag{21}$$

The final restored image R can be obtained as follows:

$$R = \frac{D_{en}}{\hat{I}_t}. \tag{22}$$

Since RI-DeNet is trained without reference images, R and $\hat{I}_t$ should be constrained separately. The total loss of RI-DeNet, $\mathcal{L}_R$, is defined as follows:

$$\mathcal{L}_R = \gamma_1\mathcal{L}_{MSE} + \gamma_2\mathcal{L}_{TV} + \gamma_3\mathcal{L}_{spa} + \gamma_4\mathcal{L}_{illu}, \tag{23}$$

where $\mathcal{L}_{MSE}$ is the fidelity following Eq. (11) to constrain the similarity between R and D_{en}. $\mathcal{L}_{TV}$ follows Eq. (16) for smoothness regularization of $\hat{I}_t$. $\mathcal{L}_{spa}$ and $\mathcal{L}_{illu}$ denote the sparsity and illumination estimation, respectively. γ_1, γ_2, γ_3 and γ_4 are trade-off parameters.

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To obtain balanced illumination in the restored image R, the sparsity loss $\mathcal{L}_{spa} = 1/HW * |R|$ is incorporated to penalize high intensity values. Due to the inherent correlation between the DoP and transmission map t in scattering imaging, we use this prior to provide physical constraints for the estimation of degraded illumination. Substituting Eq. (4) into Eq. (6), the updated transmission map t is given as follows:

$$t = 1 - \frac{S}{DoP_B \cdot B_\infty} \cdot DoP, \tag{24}$$

where DoP_B and B_∞ are global constants estimated from the selected backscattered light region. In Eq. (24), t depends on the observed intensity $S(x, y)$ and the estimates of DoP_B and B_∞, making it sensitive to local noise and estimation errors under low-light underwater scenes. Therefore, instead of directly using Eq. (24) to obtain an exact transmission estimate, we introduce a learnable parameter c to construct a stable approximation that preserves the negative correlation between transmission and DoP, as follows:

$$t = 1 - c \cdot DoP, \tag{25}$$

where the pixel-wise DoP retains the spatial variation, while c adjusts the overall strength of this relationship for the current scene. Considering that the DoP values of underwater low-light images are predominantly distributed within $[0, 0.5]$, we set the learnable scaling coefficient c within $[1, 2]$, which allows the strength of the polarization prior to be adjusted while keeping most values of $t = 1 - c \cdot DoP$ within the valid range $[0, 1]$. To ensure the physical validity of transmission, values of t below zero are set to zero during implementation.

Based on this simplified transmission relation, we define the illumination loss $\mathcal{L}_{illu}$ for estimating the degraded illumination $\hat{I}_t$, as follows:

$$\mathcal{L}_{illu} = \|\hat{I}_t - \mathcal{K} \cdot t\|_2^2, \tag{26}$$

where $\mathcal{K}$ is a global reference value for illumination estimation. Since the image intensity is normalized to [0,1], $\mathcal{K} = 0.5$ represents a moderate illumination level, consistent with common practice in brightness and exposure adjustment methods [19,60].

4. Experiments

4.1. Implementation details

Our proposed UPTD-Net comprises two stages: BD-DeNet for descattering and RI-DeNet for illumination recovery. BD-DeNet includes two branches, D-Net and B-Net. D-Net adopts a multi-scale convolutional architecture with 32 output channels and kernel sizes of 3×3, 5×5, 7×7, 5×5, and 3×3 to capture features at different scales. Since backscattered light typically exhibits a low-frequency spatial distribution, B-Net employs a uniform structure with 16 channels and 5×5 kernels to effectively capture backscattered light patterns. For RI-DeNet, the context window size is set to 5×5. The context-window and coordinate branches each contain two layers, and their outputs are fed into two separate output layers for prediction.

For each input polarization image set, UPTD-Net is independently initialized and optimized in an unsupervised manner using only the observed polarization measurements, without paired clean images, external training sets, or pre-trained models. The resulting parameters are not shared across inputs. We adopt a two-stage optimization strategy. In the first stage, D-Net and B-Net are jointly optimized for 300 epochs with learning rates of 0.001 and 0.0001, respectively. For the laboratory data, the trade-off parameters λ_1–λ_6 are set to 1, 1, 1, 5, 1, and 1, respectively. For the pool data, only λ_4 which controls the low-frequency constraint on the backscatter estimate, is increased from 5 to 10 to account for stronger backscattered light over longer acquisition distances. Upon completion of the first stage, BD-DeNet is kept fixed, and the enhanced descattering result D_{en} is passed to the second stage. In the second stage, RI-DeNet is optimized for 100 epochs with a learning rate of 1×10^{-5}. The regularization parameters γ_1–γ_4 are set to 1, 20, 5, and 10, respectively. The learnable parameter c in (25) is set with an independent learning rate of 0.01 and constrained to the interval [1, 2] during training. Mixed-precision training with gradient scaling is used to improve computational efficiency and memory utilization. We implemented our framework on an NVIDIA RTX 4060 GPU using PyTorch.

4.2. Experiment settings

Datasets: We divide our data into laboratory and pool datasets. The laboratory dataset comprises 30 scenes captured under four illumination levels in a water tank, in which milk was added to simulate turbidity, and corals and shells were used as imaging targets, yielding 120 raw DoFP images. The pool dataset contains 300 raw DoFP images acquired in a $20\text{m} \times 50\text{m}$ testing pool in Tianjin, covering calibration boards, ROVs, and metal models at various distances. The laboratory and pool images were captured using a Daheng MER2-503-23GM-P POL camera and a LUCID PHX050 S-PC camera, respectively, both at a resolution of 2448×2048 pixels. Each raw DoFP image consists of a repeated 2×2 micropolarizer pattern corresponding to polarization directions of 0°, 45°, 90°, and 135°. The raw image is downsampled according to this pattern to obtain four polarization sub-images, each with a resolution of 1224×1024 pixels.

Compared Methods: We compare our method with eight methods: three traditional methods, NHFC[19], HLRP[61], and PDS[12], and five deep learning methods, PUIE[16], SCNet[62], USUIR[52], TACL[14], and SMDR-IS[63]. NHFC and PDS are polarization-based underwater image restoration methods. Since NHFC did not provide source code, we reproduced it based on the paper's principles and parameter settings. For other methods, we directly use the source code provided by the authors. As our images are grayscale, we replicate the channel to create RGB inputs for each method.

Evaluation Metrics: Since our dataset contains only underwater polarized images without clean reference images, full-reference metrics such as PSNR and SSIM are unavailable. Therefore, we adopt five no-reference image quality metrics: EME[58], AG[64], UIQM[65], FADE[66], and NIQE[67]. Higher values of EME, AG, and UIQM generally indicate better contrast, stronger gradient responses, and higher image quality. Lower FADE and NIQE values indicate better descattering and naturalness. UIQM consists of colorfulness (UICM), sharpness (UISM), and contrast (UIConM). For grayscale images, UICM is excluded, and UIQM is computed using only sharpness and contrast terms.

4.3. Visual comparisons

Fig. 5 shows restoration results under different turbidity levels in laboratory scenes, where the first two rows correspond to low turbidity and the last two to high turbidity. Fig. 6 shows detailed comparisons of the corresponding results. Fig. 5 demonstrates that our method achieves significant descattering with improved contrast and brightness. HLRP, PUIE, and USUIR improve brightness but show insufficient descattering, with blurred textures and poor contrast. Furthermore, HLRP and USUIR produce overexposure artifacts in white regions. NHFC, PDS, SCNet, and SMDR-IS produce overall dark results with insufficient contrast under low-light underwater conditions. NHFC and SMDR-IS bury some object information in dark regions, making object identification challenging. Although TACL achieves better balance of contrast and brightness, it remains inferior to our proposed method in preserving object details and enhancing textures.

Fig. 7 shows visual comparisons under varying illumination levels. At low illumination levels, NHFC, PDS, SCNet, and SMDR-IS produce dark results with obscured details. HLRP, PUIE, USUIR, and TACL show improvements but suffer from noise and poor contrast. As illumination increases, some competing methods recover more scene information, but residual scattering, overexposure, and incomplete restoration remain

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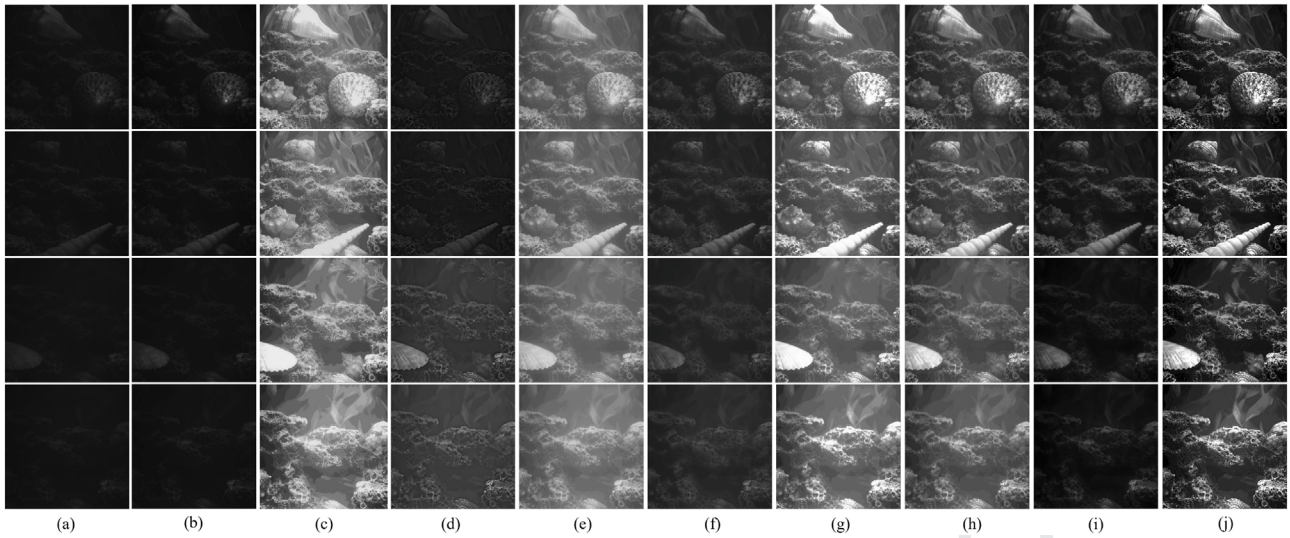


Fig. 5. Visual comparison of different methods on the laboratory dataset. The first two rows correspond to results from low-turbidity water, and the last two rows correspond to results from high-turbidity water. (a) Original. (b) NHFC[19]. (c) HLRP[61]. (d) PDS[12]. (e) PUIE[16]. (f) SCNet[62]. (g) USUIR[52]. (h) TACL[14]. (i) SMDR-IS[63]. (j) UPTD-Net.

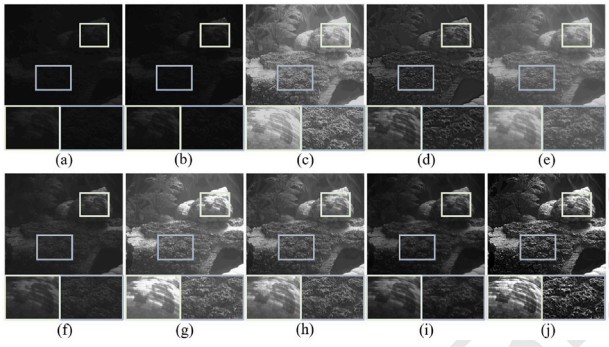


Fig. 6. Visual comparisons of detail enhancement. (a) Original. (b) NHFC[19]. (c) HLRP[61]. (d) PDS[12]. (e) PUIE[16]. (f) SCNet[62]. (g) USUIR[52]. (h) TACL[14]. (i) SMDR-IS[63]. (j) UPTD-Net.

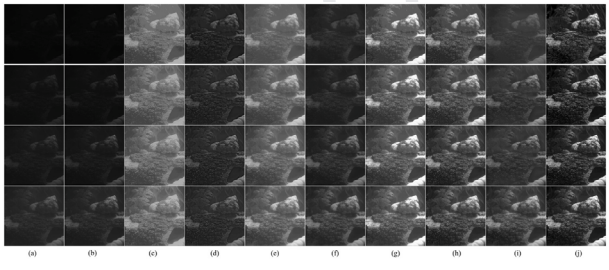


Fig. 7. Visual comparison under varying illumination conditions on the laboratory dataset. Illumination intensity increases from top to bottom rows. (a) Original. (b) NHFC[19]. (c) HLRP[61]. (d) PDS[12]. (e) PUIE[16]. (f) SCNet[62]. (g) USUIR[52]. (h) TACL[14]. (i) SMDR-IS[63]. (j) UPTD-Net.

evident. By comparison, UPTD-Net demonstrates stable restoration performance under different illumination conditions, achieving enhanced image brightness while descattering and preserving fine details.

Fig. 8 reports results for moving ROVs and metal models. NHFC and SMDR-IS exhibit limited descattering, but overall brightness remains dark with poor visual quality. HLRP and USUIR improve brightness at the cost of obvious overexposure, which causes information loss on robot surfaces. SCNet and SMDR-IS produce noticeable artifacts around the edges of the underwater robot. PDS performs well in descattering and brightness enhancement, but remains inferior to our method in contrast improvement and descattering capability.

Fig. 9 shows comparisons of the calibration-board dataset captured at different distances. NHFC and PDS fail to recover the images, with the object information submerged in darkness. HLRP, PUIE, and SMDR-IS achieve brightness enhancement, but images remain shrouded by white scattering with low clarity and contrast. Furthermore, SCNet and SMDR-IS exhibit obvious artifacts around objects. USUIR, TACL, and UPTD-Net show better descattering and illumination enhancement than other methods. However, the enlarged regions reveal that USUIR and TACL suffer from stronger noise and weaker local details, whereas UPTD-Net preserves clearer structures and more stable contrast.

4.4. Quantitative comparisons

We conducted quantitative comparisons using five no-reference image quality evaluation metrics on our datasets. Table 1 presents the average values of metrics for various methods on the laboratory dataset. Our method achieves optimal values in EME, UIQM, and FADE. From the quantitative comparison in Fig. 10, our method shows absolute advantages in the EME and UIQM distributions. It also achieves the best descattering performance for the FADE distribution, demonstrating significant advantages in clarity and contrast. Our method ranks second in both AG and NIQE. Since AG is sensitive to both image details and noise, and NIQE is based on natural image statistics, these results should be considered together with the best EME, UIQM, and FADE values, which demonstrate the balanced restoration performance of UPTD-Net.

As shown in Table 2, our method achieves the best EME and FADE results on both datasets, indicating superior contrast enhancement and descattering performance. For UIQM, our method ranks first in Scene 1 and second in Scene 2, confirming effective handling of underwater-specific degradation factors. However, our method shows relatively modest performance in AG, because our method prioritizes balanced enhancement over aggressive edge sharpening, which may introduce over-enhancement artifacts. NIQE tends to favor natural image statistics that may not fully align with the enhanced characteristics our method produces for underwater scenes. Despite these trade-offs, the overall metric gains indicate that our method effectively enhances underwater low-light image quality while preserving visual naturalness.

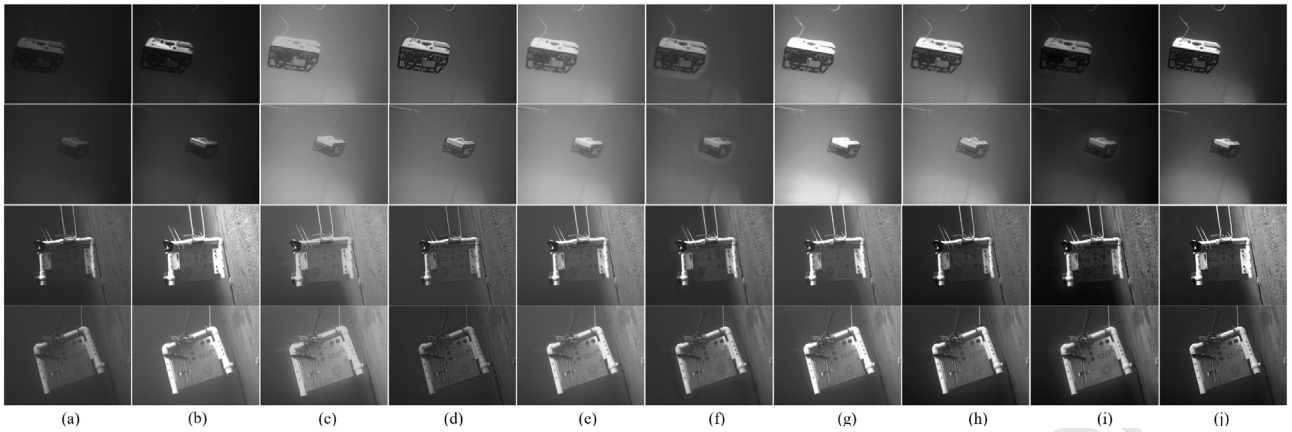


Fig. 8. Visual comparison with different methods in a large-scale testing pool. (a) Original. (b) NHFC[19]. (c) HLRP[61]. (d) PDS[12]. (e) PUIE[16]. (f) SCNet[62]. (g) USUIR[52]. (h) TACL[14]. (i) SMDR-IS[63]. (j) UPTD-Net.

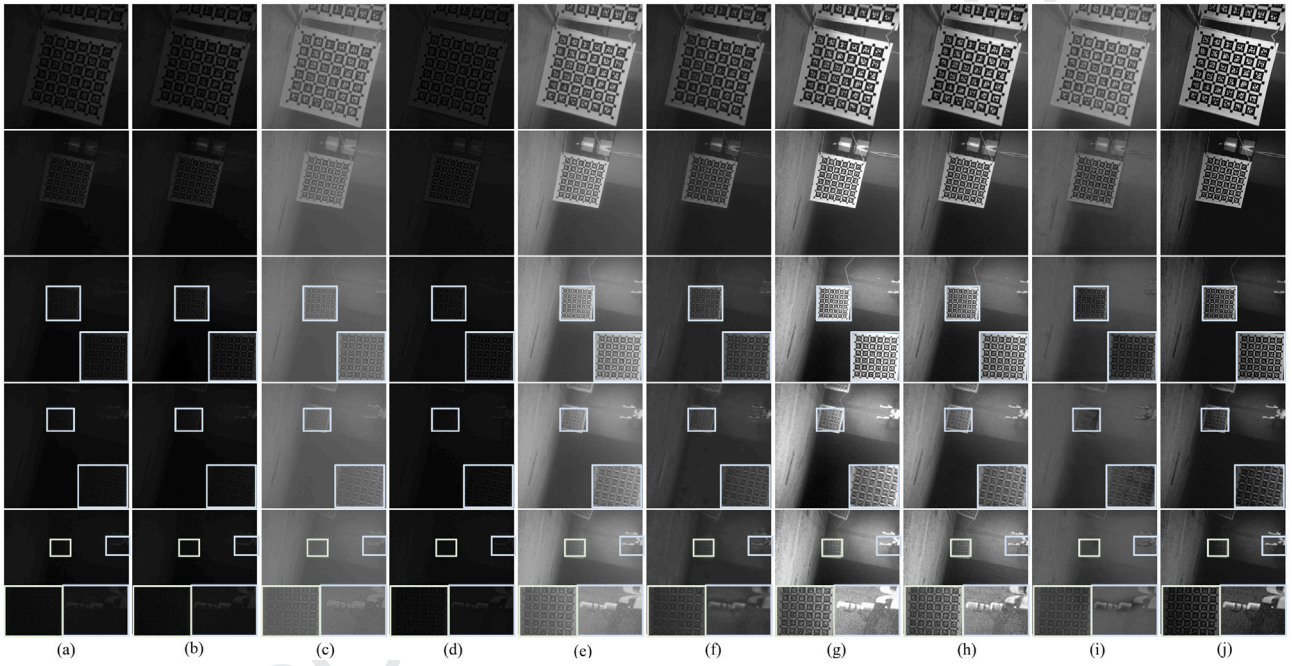


Fig. 9. Visual comparison with different methods on the calibration board dataset at different distances in a large-scale testing pool. (a) Original. (b) NHFC[19]. (c) HLRP[61]. (d) PDS[12]. (e) PUIE[16]. (f) SCNet[62]. (g) USUIR[52]. (h) TACL[14]. (i) SMDR-IS[63]. (j) UPTD-Net.

Table 1
Quantitative comparisons of different methods. The best result in each case is shown in bold, while the second-best result is highlighted in blue.

Method	EME↑	AG↑	UIQM↑	FADE↓	NIQE↓
Original	1.452	3.841	0.941	2.350	6.353
NHFC [19]	2.147	5.775	1.284	2.068	6.015
PDS [12]	3.241	17.361	1.159	1.395	4.427
HLRP [61]	3.146	**39.470**	1.776	1.146	6.628
PUIE [16]	1.334	20.306	0.920	3.003	4.063
SCNet [62]	2.071	12.714	1.050	1.903	4.843
USUIR [52]	2.263	28.371	1.689	1.979	4.179
TACL [14]	3.077	34.330	1.529	1.318	**3.183**
SMDR-IS [63]	3.352	16.344	2.059	1.489	4.053
Ours	**8.077**	37.668	**4.799**	**0.914**	3.850

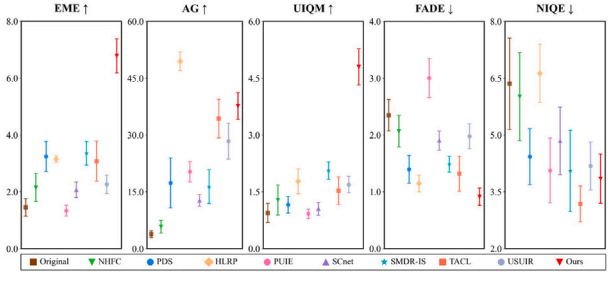


Fig. 10. Quantitative comparison of five metrics on the laboratory dataset. The nodes represent average values, and the line lengths indicate the range of fluctuations.

Table 2
Quantitative comparisons of different methods on two datasets. The best and second-best results are shown in bold and blue, respectively.

Method	Scene 1					Scene 2				
	EME↑	AG↑	UIQM↑	FADE↓	NIQE↓	EME↑	AG↑	UIQM↑	FADE↓	NIQE↓
Original	5.392	9.165	1.541	0.759	11.138	0.991	3.687	0.784	2.213	13.090
NHFC [19]	9.848	13.486	2.381	0.477	10.719	2.597	11.297	1.279	1.371	11.499
PDS [12]	7.015	11.671	1.351	0.624	9.085	2.129	22.091	0.882	1.063	8.791
HLRP [61]	2.434	30.293	0.645	1.248	10.609	1.441	21.027	0.744	1.538	12.773
PUIE [16]	3.405	39.973	1.264	0.814	8.696	0.930	14.233	0.750	2.217	10.727
SCNet [62]	4.485	24.158	1.255	0.802	10.722	0.972	7.835	0.665	2.263	11.409
SMDR-IS [63]	3.745	29.677	1.117	0.938	10.018	4.100	9.669	**2.337**	1.191	11.655
TACL [14]	5.934	57.604	2.299	0.534	**4.307**	1.415	20.951	1.114	1.598	**5.566**
USUIR [52]	10.190	**77.610**	2.922	0.426	8.545	1.842	24.310	1.359	1.159	9.269
Ours	**28.077**	50.614	**7.298**	**0.309**	7.882	**4.743**	**26.820**	1.646	**1.042**	8.945

Table 3
Quantitative comparisons of the ablation experiments. The best result in each case is shown in bold.

Setting	BD-DeNet	PDO	RI-DeNet		EME↑	AG↑	UIQM↑	FADE↓	NIQE↓
			w/o $\mathcal{L}_{\text{illu}}$	w/ $\mathcal{L}_{\text{illu}}$					
(a)	×	×	–	–	1.452	3.841	0.941	2.350	6.353
(b)	✓	×	–	–	6.391	15.467	3.860	0.993	4.436
(c)	✓	✓	–	–	7.639	21.335	4.377	0.934	4.268
(d)	✓	✓	✓	–	7.482	34.077	4.505	0.940	**3.681**
(e)	✓	✓	–	✓	**8.077**	**37.668**	**4.799**	**0.914**	3.850

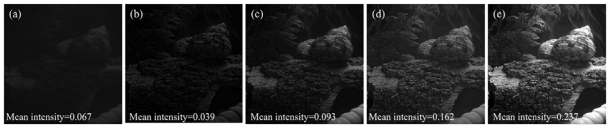


Fig. 11. Visual results of the progressive ablation study. (a) Original image. (b) BD-DeNet output. (c) PDO-enhanced result. (d) RI-DeNet output without $\mathcal{L}_{\text{illu}}$. (e) Complete UPTD-Net result. The mean intensity is reported for each image.

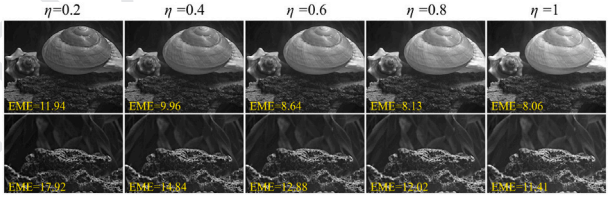


Fig. 12. Visual comparison for different values of η. As the value of η increases, the image brightness improves, but the EME value decreases, indicating reduced image contrast.

4.5. Ablation study

To validate the effectiveness of each component in UPTD-Net, we conduct ablation studies with five settings: (a) the original image, (b) the output of BD-DeNet, and (c) the result after PDO enhancement. (d) is the result of RI-DeNet without $\mathcal{L}_{\text{illu}}$, while (e) is the result of the complete model. Table 3 shows the quantitative results. BD-DeNet significantly improves EME and FADE, demonstrating its effectiveness for scattering removal. PDO improves all five metrics based on the initial descattering result. With the complete model, EME, AG, and UIQM increase from 7.639, 21.335, and 4.377 to 8.077, 37.668, and 4.799, while FADE and NIQE decrease from 0.934 and 4.268 to 0.914 and 3.850. These results show that RI-DeNet further improves the low-light image after descattering and detail enhancement. The comparison between settings (d) and (e) further demonstrates the contribution of $\mathcal{L}_{\text{illu}}$ to illumination recovery in RI-DeNet.

Fig. 11 shows the corresponding visual results. BD-DeNet effectively removes scattering and improves image contrast, while PDO enhances image details and brightness. After RI-DeNet, dark regions are better recovered and previously obscured structures become clearer. The mean intensity increases from 0.093 after PDO to 0.237 with the complete RI-DeNet, showing its effectiveness in illumination recovery.

We conducted an experimental analysis on the value of parameter η in (20), as shown in Fig. 12. Smaller η values produce darker images with higher contrast, while larger η values produce brighter images with reduced contrast. To balance brightness and contrast, we recommend η in the range of 0.4 to 0.8. In this paper, we set η to 0.8.

4.6. Parameter sensitivity and convergence

Loss weight sensitivity. We evaluate the sensitivity of the loss weights by varying one weight at a time while keeping the others at their default values. For λ_1–λ_3, λ_5–λ_6, and γ_1, we test $\{0.25, 0.5, 1, 2, 4\}$. The tested values for λ_4, γ_2, γ_3, and γ_4 are $\{1, 2, 5, 10, 20\}$, $\{5, 10, 20, 40, 80\}$, $\{1, 2, 5, 10, 20\}$, and $\{2, 5, 10, 20, 40\}$, respectively. To combine metrics with different scales and optimization directions, we define two normalized scores:

$$S_{\uparrow} = \frac{1}{3}\left(\frac{\text{EME}}{\text{EME}_0} + \frac{\text{AG}}{\text{AG}_0} + \frac{\text{UIQM}}{\text{UIQM}_0}\right),$$
$$S_{\downarrow} = \frac{1}{2}\left(\frac{\text{FADE}_0}{\text{FADE}} + \frac{\text{NIQE}_0}{\text{NIQE}}\right). \tag{27}$$

Here, EME_0, AG_0, UIQM_0, FADE_0, and NIQE_0 denote the metric values obtained using the default loss weights. With this definition, higher values indicate better performance, and the default configuration corresponds to $S_{\uparrow} = S_{\downarrow} = 1$.

As shown in Fig. 13, most parameters exhibit relatively small variations over a wide range with low parameter sensitivity. The effects of λ_1 and λ_3 are more noticeable. Increasing λ_1 keeps D closer to the degraded image S, which limits scattering removal and lowers both scores. Increasing λ_3 strengthens the reconstruction relation $D + B \approx S$ and improves both scores when the other weights are fixed. The adopted setting $\lambda_1 = \lambda_3 = 1$ reflects the joint effect of the fidelity and reconstruction losses. For RI-DeNet, a smaller γ_3 allows stronger enhancement, whereas

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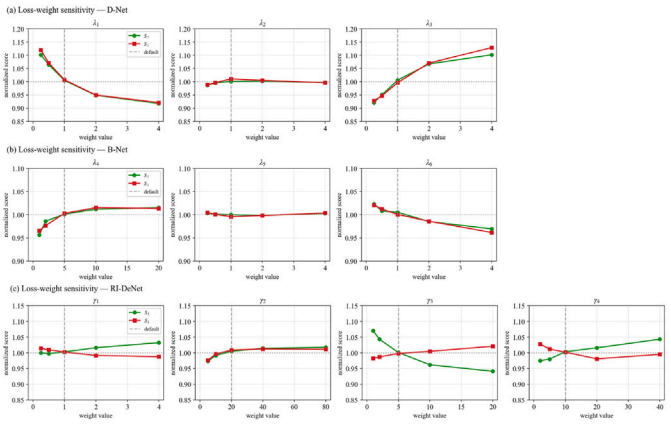


Fig. 13. Sensitivity analysis of the loss weights. Each weight is varied individually while the remaining weights are fixed at their default values. Higher values of $S_{\uparrow}$ and $S_{\downarrow}$ indicate better performance.

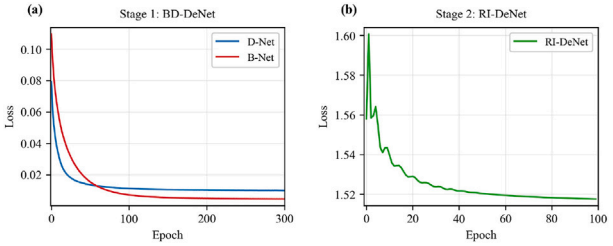


Fig. 14. Optimization convergence curves of UPTD-Net. (a) Losses of D-Net and B-Net. (b) Loss of RI-DeNet.

a larger γ_3 better suppresses excessive enhancement. The adopted value $\gamma_3 = 5$ reflects the trade-off between enhancement strength and image naturalness. Increasing γ_4 strengthens the illumination loss and improves the normalized scores when tested independently. However, an excessively large γ_4 will reduce the relative effects of the other loss terms, and $\gamma_4 = 10$ maintains a balanced contribution during joint optimization. Since each curve varies only one weight, its individual optimum may not be optimal for the complete loss. The default weights were therefore selected by considering all loss terms and evaluation metrics together.

Optimization convergence. Fig. 14 shows the loss curves of the two optimization stages. The losses of D-Net and B-Net decrease rapidly during the early epochs and gradually stabilize over 300 epochs. Although the curves become relatively flat earlier, we retain 300 epochs to obtain a stable scattering decomposition for the subsequent stage and reduce error propagation. RI-DeNet exhibits a brief initial fluctuation, followed by a steady decrease, and converges within 100 epochs. These curves indicate stable optimization behavior in both stages.

4.7. Efficiency analysis

Table 4 compares the model parameters, pretraining requirements, and computational cost of different methods. UPTD-Net contains only 0.262 M parameters, including 0.102 M, 0.020 M, and 0.139 M for D-Net, B-Net, and RI-DeNet, respectively. Despite its two-stage architecture, its parameter count is slightly higher than USUIR and lower than other learning-based methods. Since the inference time of pretrained methods excludes offline training cost, whereas UPTD-Net performs image-specific unsupervised optimization without pretraining, their computational costs are not directly comparable. We therefore report the forward time of UPTD-Net (0.042 s) for inference-time comparison and additionally provide its complete optimization time of 13.1 s per image. Overall, UPTD-Net remains lightweight with an acceptable cost for image-specific optimization.

Table 4
Comparison of model complexity and computational efficiency. Runtimes are measured on 450 × 450 images using an NVIDIA GeForce RTX 4060 Laptop GPU.

Method	Params (M)	Pretraining	Inference time (s)
NHFC [19]	N/A	No	2.341
HLRP [61]	N/A	No	0.636
PDS [12]	N/A	No	1.910
PUIE [16]	1.401	Yes	0.741
SCNet [62]	0.818	Yes	0.048
USUIR [52]	0.226	Yes	0.042
TACL [14]	11.378	Yes	0.185
SMDR-IS [63]	12.557	Yes	0.086
UPTD-Net	0.262	No	0.042
D-Net	0.102	—	0.008
B-Net	0.020	—	0.004
RI-DeNet	0.139	—	0.028

Table 5
Consistency of RGB-based methods between native RGB and replicated grayscale inputs.

Method	PSNR	SSIM	MAE
PUIE	34.95	0.994	0.0155
SCNet	41.57	0.996	0.0063
USUIR	36.35	0.997	0.0108
TACL	31.16	0.989	0.0221
SMDR-IS	27.63	0.917	0.0510

4.8. Grayscale input adaptation analysis

Since some compared methods were originally trained on color images, replicating grayscale inputs to three channels may affect their performance. Therefore, we evaluated this effect using the data in Fig. 15 by comparing the outputs obtained from native RGB and replicated grayscale inputs, where Y = 0.299R + 0.587G + 0.114B. As shown in Table 5, most methods exhibit high output consistency, whereas SMDR-IS shows a larger variation. This indicates that the effect of grayscale replication is method dependent but limited for most evaluated methods. Since the main comparisons focus on scattering removal and illumination recovery under the same grayscale input condition, the variations are considered acceptable for assessing the relative restoration performance of different methods.

4.9. Generalization performance

To further explore the transferability of the proposed two-stage decoupling strategy, we conduct additional experiments on color polarized images, non-polarized underwater images, and hazy images. These experiments serve as auxiliary validation of the proposed framework under different input settings.

Color Polarized Underwater Imaging: Our proposed method focuses on addressing the coupled degradation of scattering and low illumination in underwater scenarios, without designing a specific module for color restoration. For color polarized data, our decoupling network processes the R, G, and B channels independently, followed by a simple Gray World white balance. Although no complex color-specific designs are introduced, our method still achieves superior visual quality compared to other algorithms, as shown in Fig. 15. Driven by the exceptional descattering and illumination recovery capabilities of our network, even a simple color calibration is sufficient to yield outstanding color restoration results.

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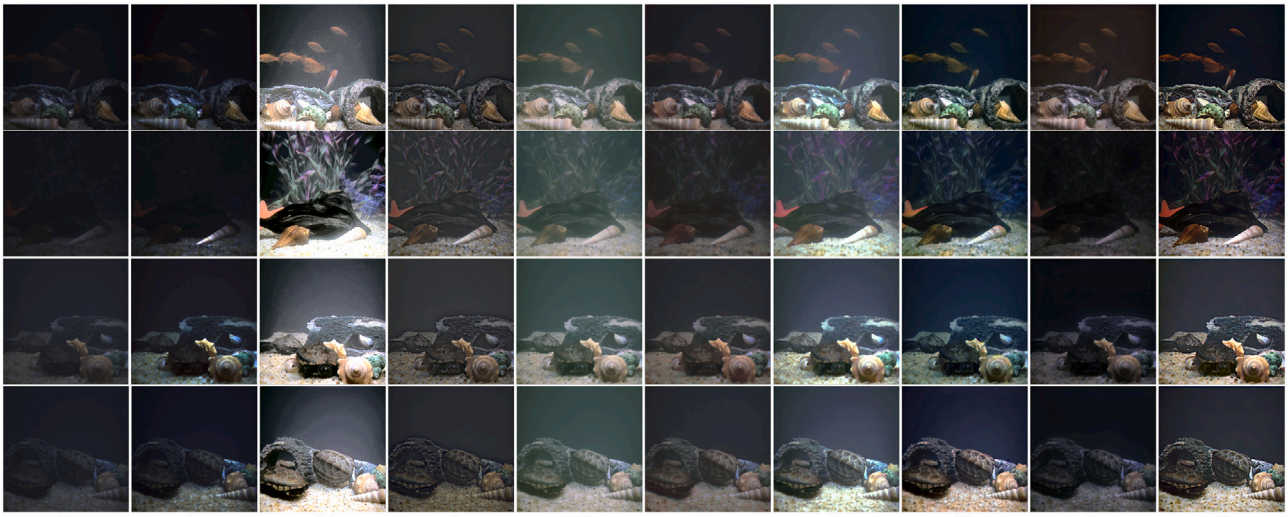


Fig. 15. Visual comparison of color polarized underwater images. (a) Original. (b) NHFC[19]. (c) HLRP[61]. (d) PDS[12]. (e) PUIE[16]. (f) SCNet[62]. (g) USUIR[52]. (h) TACL[14]. (i) SMDR-IS[63]. (j) UPTD-Net.

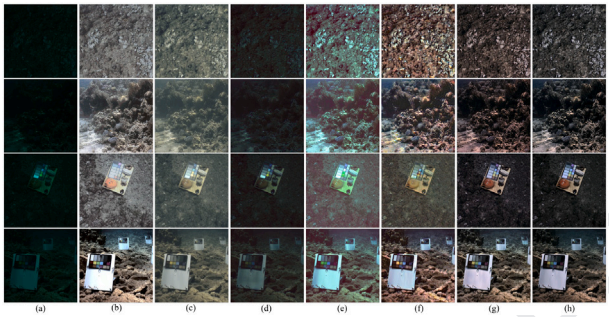


Fig. 16. Visual comparison of non-polarized low-light underwater color images. (a) Original image. (b) HLRP[61]. (c) PUIE[16]. (d) SCNet[62]. (e) USUIR[52]. (f) TACL[14]. (g) SMDR-IS[63]. (h) UPTD-Net.

Non-polarized Underwater Low-light Imaging: Although designed for underwater grayscale polarized data, our method applies effectively to non-polarized underwater low-light images. In order to achieve this, the RGB images are converted to HSV space, with the V component extracted as network input. The enhanced V component is then recombined with the original H and S components. The gray world white balance algorithm is employed for color correction. We evaluate UPTD-Net on the Sea-Thru dataset [68], and compare it with six non-polarized restoration methods: HLRP, PUIE, SCNet, SMDR-IS, TACL and USUIR. The results presented in Fig. 16 demonstrate that the proposed method produces excellent enhancement effects on non-polarized underwater low-light images, exhibiting strong adaptability to underwater low-light environments.

Hazy Image: To further validate generalization capability, we evaluate our method on image dehazing tasks. We select images from the RESIDE [69] dataset for testing, with the results shown in Fig. 17. The same processing pipeline as that for underwater RGB images is adopted, except that color correction is omitted. Results demonstrate significant descattering effects on hazy images under both normal and low-light conditions. Enhancement of low-light hazy images highlights the method's superiority in joint descattering and low-light enhancement, achieving simultaneous optimization of scattering removal and illumination enhancement.

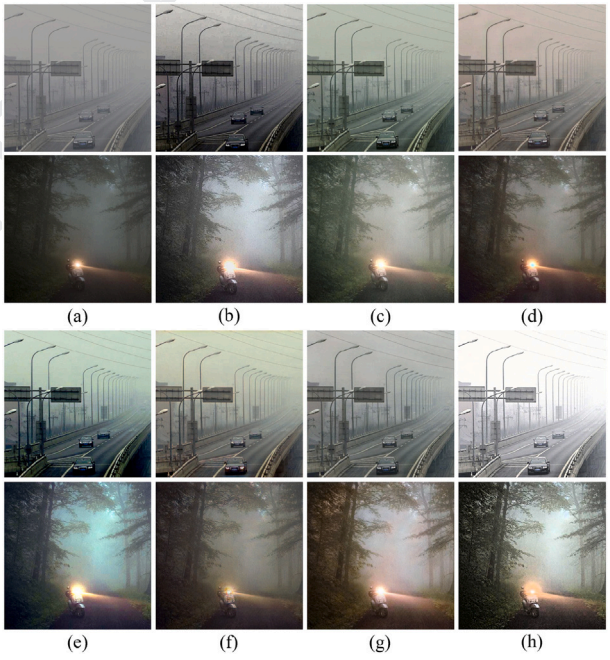


Fig. 17. Visual comparison of hazy images. (a) Original image. (b) HLRP[61]. (c) PUIE[16]. (d) SCNet[62]. (e) USUIR[52]. (f) TACL[14]. (g) SMDR-IS[63]. (h) UPTD-Net.

5. Conclusion

This paper presents UPTD-Net, an unsupervised two-stage framework for restoring underwater images degraded by scattering and low illumination. The restoration process is divided into scattering removal and illumination recovery without paired data or pretraining. Polarization information is incorporated through spatial weighting and a soft physical reference to refine the descattered image and optimize the estimation of degraded illumination, respectively. Quantitative comparisons and progressive ablation results support the effectiveness of these two uses of polarization information.

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This paper mainly focuses on decoupling scattering and illumination degradation in underwater low-light conditions, without introducing a dedicated module for color degradation. Nevertheless, generalization experiments show that the proposed method still achieves satisfactory results on both color polarized images and real-world color images. In future work, we will further explore restoration methods for underwater RGB images and polarization-aware enhancement strategies, and extend the framework to color polarized image restoration to further investigate the synergy between polarization information and deep learning.

CRediT authorship contribution statement

Xiaotong Fei: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Linghao Shen:** Writing – review & editing, Visualization, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Yizhao Huang:** Validation, Software, Investigation, Data curation. **Haofeng Hu:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Dagong Jia:** Supervision, Resources, Project administration, Formal analysis. **Binglu Wang:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.optlastec.2026.116479.

Data availability

Data will be made available on request.

References

[1] B. Li, Z. Chen, L. Lu, P. Qi, L. Zhang, Q. Ma, H. Hu, J. Zhai, X. Li, Cascaded frameworks in underwater optical image restoration, Inf. Fusion 117 (2025) 102809, https://doi.org/10.1016/j.inffus.2024.102809

[2] J. Guan, M. Ma, Y. Huo, Underwater polarimetric dark channel prior descattering, Opt. Laser Technol. 175 (2024) 110864, https://doi.org/10.1016/j.optlastec.2024.110864

[3] Y. Li, Y. Chen, J. Zhang, Y. Li, H. Tang, X. Fu, A robust underwater polarization image recovery based on angle of polarization with low-rank and sparse decomposition, Opt. Laser Technol. 181 (2025) 111669, https://doi.org/10.1016/j.optlastec.2024.111669

[4] B. Hu, M. Shan, L. Liu, Y. Xie, Z. Li, Prior-free underwater image restoration based on dual polarization parameters, Opt. Laser Technol. 192 (2025) 114053, https://doi.org/10.1016/j.optlastec.2025.114053

[5] M. Abdullah-Al-Wadud, M.H. Kabir, M.A. Akber Dewan, O. Chae, A dynamic histogram equalization for image contrast enhancement, IEEE Trans. Consum. Electron. 53 (2007) 593–600, https://doi.org/10.1109/TCE.2007.381734

[6] C. Li, J. Guo, R. Cong, Y. Pang, B. Wang, Underwater image enhancement by dehazing with minimum information loss and histogram distribution prior, IEEE Trans. Image Process. 25 (2016) 5664–5677, https://doi.org/10.1109/TIP.2016.2612882

[7] A. Mishra, M. Gupta, P. Sharma, Enhancement of underwater images using improved CLAHE, in: Proc. Int. Conf. Adv. Comput. Telecommun, 2018, pp. 1–6.

[8] A. Galdran, D. Pardo, A. Picón, A. Alvarez-Gila, Automatic red-channel underwater image restoration, J. Vis. Commun. Image Represent. 26 (2015) 132–145, https://doi.org/10.1016/j.jvcir.2014.11.006

[9] D. Berman, D. Levy, S. Avidan, T. Treibitz, Underwater single image color restoration using haze-lines and a new quantitative dataset, IEEE Trans. Pattern Anal. Mach. Intell. 43 (2021) 2822–2837, https://doi.org/10.1109/TPAMI.2020.2977624

[10] I. Omer, M. Werman, Color lines: image specific color representation, in: Proc. IEEE Conf. Comput. vis. Pattern Recognit, 2, 2004, pp. 2, https://doi.org/10.1109/CVPR.2004.1315267

[11] K. He, J. Sun, X. Tang, Single image haze removal using dark channel prior, IEEE Trans. Pattern Anal. Mach. Intell. 33 (2011) 2341–2353, https://doi.org/10.1109/TPAMI.2010.168

[12] L. Shen, M. Reda, X. Zhang, Y. Zhao, S.G. Kong, Polarization-driven solution for mitigating scattering and uneven illumination in underwater imagery, IEEE Trans. Geosci. Remote Sens. 62 (2024) 1–15, https://doi.org/10.1109/TGRS.2024.3358828

[13] Y. Li, X. Sun, J. Zhang, Y. Chen, Y. Li, X. Fu, Underwater polarization de-scattering method with dynamic contrast constraint and adaptive genetic algorithm, Opt. Laser Technol. 183 (2025) 112375, https://doi.org/10.1016/j.optlastec.2024.112375

[14] R. Liu, Z. Jiang, S. Yang, X. Fan, Twin adversarial contrastive learning for underwater image enhancement and beyond, IEEE Trans. Image Process. 31 (2022) 4922–4936, https://doi.org/10.1109/TIP.2022.3190209

[15] L. Peng, C. Zhu, L. Bian, U-shape transformer for underwater image enhancement, IEEE Trans. Image Process. 32 (2023) 3066–3079, https://doi.org/10.1109/TIP.2023.3276332

[16] Z. Fu, W. Wang, Y. Huang, X. Ding, K.-K. Ma, Uncertainty inspired underwater image enhancement, in: Proc. Eur. Conf. Comput. vis, Springer, 2022, pp. 465–482, https://doi.org/10.1007/978-3-031-19797-0_27

[17] L. Shen, H. Xia, X. Zhang, Y. Zhao, N. Li, S.G. Kong, B. Wang, Z. Li, U^2pnet: an unsupervised underwater image-restoration network using polarization, IEEE Trans. Cybern. 54 (2024) 5164–5177, https://doi.org/10.1109/TCYB.2024.3365693

[18] Y.Y. Schechner, Y. Averbuch, Regularized image recovery in scattering media, IEEE Trans. Pattern Anal. Mach. Intell. 29 (2007) 1655–1660, https://doi.org/10.1109/TPAMI.2007.1141

[19] Y. Chen, Y. Li, Y. Wang, Z. Mi, Y. Wang, X. Fu, Robust polarization-based underwater image enhancement method using anchor brightness adaptation, Opt. Lasers Eng. 169 (2023) 107737, https://doi.org/10.1016/j.optlaseng.2023.107737

[20] L. Shen, L. Zhang, P. Qi, X. Zhang, X. Li, Y. Huang, Y. Zhao, H. Hu, Polarimetric binocular three-dimensional imaging in turbid water with multi-feature self-supervised learning, Photonix 6 (2025), https://doi.org/10.1186/s43074-025-00185-4

[21] Q. Ma, X. Li, B. Li, Z. Zhu, J. Wu, F. Huang, H. Hu, STAMF: synergistic transformer and Mamba fusion network for RGB-polarization based underwater salient object detection, Inf. Fusion 122 (2025) 103182, https://doi.org/10.1016/j.inffus.2025.103182

[22] H. Hu, L. Zhao, X. Li, H. Wang, J. Yang, K. Li, T. Liu, Polarimetric image recovery in turbid media employing circularly polarized light, Opt. Express 26 (2018) 25047–25059, https://doi.org/10.1364/OE.26.025047

[23] H. Hu, Y. Huang, X. Li, L. Jiang, L. Che, T. Liu, J. Zhai, UCRNet: underwater color image restoration via a polarization-guided convolutional neural network, Front. Mar. Sci. 9 (2022) 1031549, https://doi.org/10.3389/fmars.2022.1031549

[24] Y. Zhu, T. Zeng, K. Liu, Z. Ren, E.Y. Lam, Full scene underwater imaging with polarization and an untrained network, Opt. Express 29 (2021) 41865–41881, https://doi.org/10.1364/OE.444755

[25] Y.-T. Peng, K. Cao, P.C. Cosman, Generalization of the dark channel prior for single image restoration, IEEE Trans. Image Process. 27 (2018) 2856–2868, https://doi.org/10.1109/TIP.2018.2813092

[26] Z. Liang, X. Ding, Y. Wang, X. Yan, X. Fu, GUDCP: generalization of underwater dark channel prior for underwater image restoration, IEEE Trans. Circuits Syst. Video Technol. 32 (2022) 4879–4884, https://doi.org/10.1109/TCSVT.2021.3114230

[27] N. Carlevaris-Bianco, A. Mohan, R.M. Eustice, Initial results in underwater single image dehazing, in: OCEANS 2010 MTS/IEEE Seattle, 2010, pp. 1–8, https://doi.org/10.1109/OCEANS.2010.5664428

[28] W. Song, Y. Wang, D. Huang, D. Tjondronegoro, A rapid scene depth estimation model based on underwater light attenuation prior for underwater image restoration, in: Pacific Rim Conference on Multimedia, 2018, pp. 678–688.

[29] G. Hou, N. Li, P. Zhuang, K. Li, H. Sun, C. Li, Non-uniform illumination underwater image restoration via illumination channel sparsity prior, IEEE Trans. Circuits Syst. Video Technol. 34 (2024) 799–814, https://doi.org/10.1109/TCSVT.2023.3290363

[30] W. Zhang, P. Zhuang, H.-H. Sun, G. Li, S. Kwong, C. Li, Underwater image enhancement via minimal color loss and locally adaptive contrast enhancement, IEEE Trans. Image Process. 31 (2022) 3997–4010, https://doi.org/10.1109/TIP.2022.3177129

[31] B. Huang, T. Liu, H. Hu, J. Han, M. Yu, Underwater image recovery considering polarization effects of objects, Opt. Express 24 (2016) 9826–9838, https://doi.org/10.1364/OE.24.009826

[32] H. Wang, J. Li, H. Hu, J. Jiang, X. Li, K. Zhao, Z. Cheng, M. Sang, T. Liu, Underwater imaging by suppressing the backscattered light based on mueller matrix, IEEE Photonics J. 13 (2021) 1–6, https://doi.org/10.1109/JPHOT.2021.3094359

[33] Y. Wei, P. Han, F. Liu, X. Shao, Enhancement of underwater vision by fully exploiting the polarization information from the Stokes vector, Opt. Express 29 (2021) 22275–22287, https://doi.org/10.1364/OE.433072

[34] D. Huang, Y. Wang, W. Song, J. Sequeira, S. Mavromatis, Shallow-water image enhancement using relative global histogram stretching based on adaptive parameter acquisition, in: Proc. Int. Conf. Multimedia Model, Springer, 2018, pp. 453–465, https://doi.org/10.1007/978-3-319-73603-7_37

[35] G. Deng, A generalized unsharp masking algorithm, IEEE Trans. Image Process. 20 (2011) 1249–1261, https://doi.org/10.1109/TIP.2010.2092441

[36] X. Fu, Y. Liao, D. Zeng, Y. Huang, X.-P. Zhang, X. Ding, A probabilistic method for image enhancement with simultaneous illumination and reflectance estimation, IEEE Trans. Image Process. 24 (2015) 4965–4977, https://doi.org/10.1109/TIP.2015.2474701

[37] W. Zhang, L. Zhou, P. Zhuang, G. Li, X. Pan, W. Zhao, C. Li, Underwater image enhancement via weighted wavelet visual perception fusion, IEEE Trans. Circuits Syst. Video Technol. 34 (2024) 2469–2483, https://doi.org/10.1109/TCSVT.2023.3299314

UNCORRECTED PROOF

[38] Y. Zheng, H. Lu, J. Wang, W. Zhang, M. Guizani, High-turbidity underwater image enhancement via turbidity suppression fusion, IEEE Trans. Circuits Syst. Video Technol. (2024), https://doi.org/10.1109/TCSVT.2024.3508102
[39] Q. Qi, Y. Zhang, F. Tian, Q.J. Wu, K. Li, X. Luan, D. Song, Underwater image co-enhancement with correlation feature matching and joint learning, IEEE Trans. Circuits Syst. Video Technol. 32 (2021) 1133–1147, https://doi.org/10.1109/TCSVT.2021.3074197
[40] P. Sharma, I. Bisht, A. Sur, Wavelength-based attributed deep neural network for underwater image restoration, ACM Trans. Multimedia Comput. Commun. Appl. 19 (2023) 1–23.
[41] D. Zhang, C. Wu, J. Zhou, W. Zhang, C. Li, Z. Lin, Hierarchical attention aggregation with multi-resolution feature learning for GAN-based underwater image enhancement, Eng. Appl. Artif. Intell. 125 (2023) 106743, https://doi.org/10.1016/j.engappai.2023.106743
[42] Z. Huang, J. Li, Z. Hua, L. Fan, Underwater image enhancement via adaptive group attention-based multiscale cascade transformer, IEEE Trans. Instrum. Meas. 71 (2022) 1–18, https://doi.org/10.1109/TIM.2022.3189630
[43] X. Chu, Z. Fu, S. Yu, X. Tu, Y. Huang, X. Ding, Underwater image enhancement and super-resolution using implicit neural networks, in: Proc. IEEE Int. Conf. Image Process, 2023, pp. 1295–1299.
[44] Z. Zhu, X. Li, Q. Ma, J. Zhai, H. Hu, FDNet: Fourier transform guided dual-channel underwater image enhancement diffusion network, Sci. China Technol. Sci. 68 (2025) 1100403, https://doi.org/10.1007/s11431-024-2824-x
[45] F. Huo, B. Li, X. Zhu, Efficient wavelet boost learning-based multi-stage progressive refinement network for underwater image enhancement, in: Proc. IEEE Conf. Comput. vis. Pattern Recognit, 2021, pp. 1944–1952.
[46] M. Guan, H. Xu, G. Jiang, M. Yu, Y. Chen, T. Luo, X. Zhang, DiffWater: underwater image enhancement based on conditional denoising diffusion probabilistic model, IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 17 (2024) 2319–2335, https://doi.org/10.1109/JSTARS.2023.3344453
[47] C. Zhao, W. Cai, C. Dong, C. Hu, Wavelet-based Fourier information interaction with frequency diffusion adjustment for underwater image restoration, in: Proc. IEEE/CVF Conf. Comput. vis. Pattern Recognit, 2024, pp. 8281–8291.
[48] F. Qian, L. Ma, K. Su, Z. Zhang, Z. Chen, J. Zhang, WDUIE: wavelet-based diffusion for semi-supervised underwater image enhancement, Opt. Laser Technol. 192 (2025) 113988, https://doi.org/10.1016/j.optlastec.2025.113988
[49] B. Lin, W. Chen, X. Fan, P. Peng, Z. Guo, Transformer-based improved U-Net for high-performance underwater polarization imaging, Opt. Laser Technol. 181 (2025) 111664, https://doi.org/10.1016/j.optlastec.2024.111664
[50] X. Fan, M. Ding, T. Lv, X. Sun, B. Lin, Z. Guo, Meta-DNET-UPI: efficient underwater polarization imaging combining deformable convolutional networks and meta-learning, Opt. Laser Technol. 187 (2025) 112900, https://doi.org/10.1016/j.optlastec.2025.112900
[51] H. Liu, M. Qiu, H. Cao, Q. Ma, X. Li, H. Hu, UPI2Diff: restoring underwater polarization image and information in turbid conditions via polarization guided diffusion model, IEEE Trans. Circuits Syst. Video Technol. 36 (2026) 7615–7629, https://doi.org/10.1109/TCSVT.2026.3656897
[52] Z. Fu, H. Lin, Y. Yang, S. Chai, L. Sun, Y. Huang, X. Ding, Unsupervised underwater image restoration: from a homology perspective, in: Proc. AAAI Conf. Artif. Intell, 36, 2022, pp. 643–651, https://doi.org/10.1609/aaai.v36i1.19944
[53] B. McGlamery, A computer model for underwater camera systems, in: Ocean Optics VI, 208, 1980, pp. 221–231.
[54] J.S. Jaffe, Computer modeling and the design of optimal underwater imaging systems, IEEE J. Oceanic Eng. 15 (2002) 101–111, https://doi.org/10.1109/48.50695
[55] E.H. Land, J.J. McCann, Lightness and retinex theory, J. Opt. Soc. Am. 61 (1971) 1–11, https://doi.org/10.1364/JOSA.61.000001
[56] J. Deng, J. Zhu, H. Li, X. Zhang, F. Guo, X. Hou, Real-time underwater polarization imaging without relying on background, Opt. Lasers Eng. 169 (2023) 107721, https://doi.org/10.1016/j.optlaseng.2023.107721
[57] J. Liang, L. Ren, R. Liang, Low-pass filtering based polarimetric dehazing method for dense haze removal, Opt. Express 29 (2021) 28178–28189, https://doi.org/10.1364/OE.427629
[58] S.S. Agaian, K. Panetta, A.M. Grigoryan, Transform-based image enhancement algorithms with performance measure, IEEE Trans. Image Process. 10 (2001) 367–382, https://doi.org/10.1109/83.908502
[59] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, G. Wetzstein, Implicit neural representations with periodic activation functions, in: Adv. Neural Inf. Process. Syst, 33, Curran Associates, Inc., 2020, pp. 7462–7473.
[60] T. Chobola, Y. Liu, H. Zhang, J.A. Schnabel, T. Peng, Fast context-based low-light image enhancement via neural implicit representations, in: Proc. Eur. Conf. Comput. vis, Springer, 2024, pp. 413–430, https://doi.org/10.1007/978-3-031-73016-0_24
[61] P. Zhuang, J. Wu, F. Porikli, C. Li, Underwater image enhancement with hyper-laplacian reflectance priors, IEEE Trans. Image Process. 31 (2022) 5442–5455, https://doi.org/10.1109/TIP.2022.3196546
[62] Z. Fu, X. Lin, W. Wang, Y. Huang, X. Ding, Underwater image enhancement via learning water type desensitized representations, in: Proc. IEEE Int. Conf. Acoust, Speech, Signal Process, 2022, pp. 2764–2768, https://doi.org/10.1109/ICASSP43922.2022.9747758
[63] D. Zhang, J. Zhou, C. Guo, W. Zhang, C. Li, Synergistic multiscale detail refinement via intrinsic supervision for underwater image enhancement, in: Proc. AAAI Conf. Artif. Intell, 38, 2024, pp. 7033–7041, https://doi.org/10.1609/aaai.v38i7.28530
[64] W. Zhang, L. Dong, X. Pan, J. Zhou, L. Qin, W. Xu, Single image defogging based on multi-channel convolutional MSRCR, IEEE Access 7 (2019) 72492–72504, https://doi.org/10.1109/ACCESS.2019.2920403
[65] K. Panetta, C. Gao, S. Agaian, Human-visual-system-inspired underwater image quality measures, IEEE J. Oceanic Eng. 41 (2015) 541–551, https://doi.org/10.1109/JOE.2015.2469915
[66] L.K. Choi, J. You, A.C. Bovik, Referenceless prediction of perceptual fog density and perceptual image defogging, IEEE Trans. Image Process. 24 (2015) 3888–3901, https://doi.org/10.1109/TIP.2015.2456502
[67] A. Mittal, R. Soundararajan, A.C. Bovik, Making a “completely blind” image quality analyzer, IEEE Signal Process Lett. 20 (2012) 209–212, https://doi.org/10.1109/LSP.2012.2227726
[68] D. Akkaynak, T. Treibitz, Sea-thru: a method for removing water from underwater images, in: Proc. IEEE Conf. Comput. vis. Pattern Recognit, 2019.
[69] B. Li, W. Ren, D. Fu, D. Tao, D. Feng, W. Zeng, Z. Wang, Benchmarking single-image dehazing and beyond, IEEE Trans. Image Process. 28 (2019) 492–505, https://doi.org/10.1109/TIP.2018.2867951

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